18V 3.5A 1MHz Synchronous Rectified Step-Down DC/DC Converter

DESCRIPTION

The MT3503/MT3503A is a high efficiency, 3.5A current mode synchronous step-down DC/DC converter with a wide input voltage range from 4.5V to 18V. The device integrates lower high side and low side MOSFETs to achieve high efficiency conversion. The current mode control architecture supports fast transient response and internal compensation. The MT3503/MT3503A provides complete fault protection such as input under-voltage lockout, output short circuit protection, over current protection, and thermal shutdown. The switching frequency is internally set at 1MHz. MT3503 and MT3503A have different operation modes:

- MT3503 is automatic PSM/PWM mode.
- MT3503A is Forced PWM mode.

To improve the light load efficiency, MT3503 has the light load power saving mode (PSM) to minimize the switching loss by reducing the switching frequency. The MT3503/MT3503A is available in the TSOT23_6L package.

FEATURES

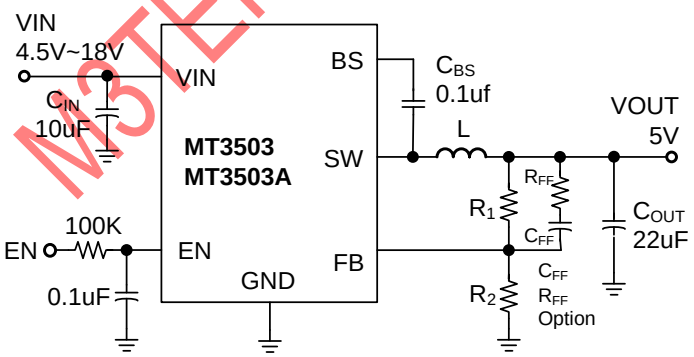
- Input Voltage Range from 4.5V to 18V
- +/- 2% 0.6V Feedback Voltage Accuracy
- Adjustable Output Voltage from 0.6V to 12V
- 1MHz Switching Frequency
- Built-in 70mΩ/30mΩ Power Switch
- Continuous Output Current up to 3.5A
- High Efficiency up to 94%
- Internal Soft-Start
- Input Under Voltage Lockout (UVLO)
- Cycle-by-Cycle Current Limit Protection
- Over-Load and Short Circuit Protection
- Thermal Shutdown Protection
- Available in a Small TSOT23_6L Package
- Pb-Free RoHS Compliant

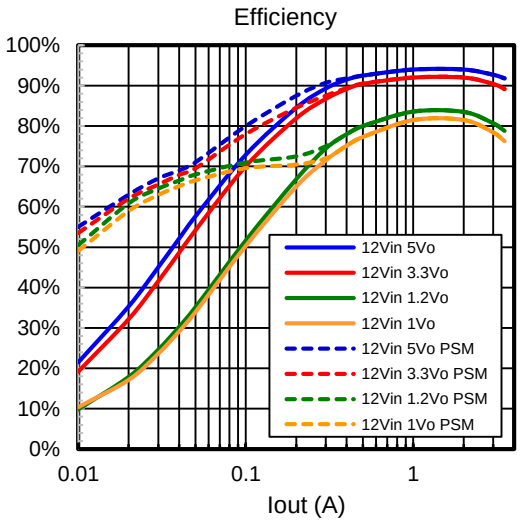
APPLICATIONS

- Wireless and DSL Card
- Portable/Handheld Device
- STB, TV, Sound Bar, MP3 Player
- Slim DVD, SSD
- Microprocessor and DSP Core Supply

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Typical Applications



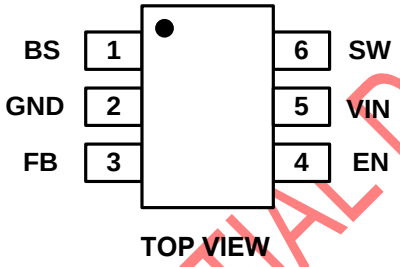


Ordering Information

Part No.	Marking	Temp. Range	Package	MOQ
MT3503NTBR	3503 YWWxx	-40°C ~+85°C	TSOT23_6L	3000/Tape & Reel
MT3503ATBR	3503A YWWxx	-40°C ~+85°C	TSOT23_6L	3000/Tape & Reel

Note: Y:Year, WW:Week, xx:Manufacture Control Code

Pin Configuration



TOP VIEW

Pin Description

Pin No.	Symbol	Description
1	BS	High Side Gate Drive Boost Input Connect a 0.1uF ceramic capacitor between BS and SW pins.
2	GND	Ground This is the reference of the ground connection for all components in the power supply.
3	FB	Voltage Feedback
4	EN	Regulator Enable Control Input, Don't float this PIN • Drive EN High Level to turn on the converter • Drive EN Low Level to turn off the converter
5	VIN	Main Input Supply Voltage
6	SW	Power Switches Node

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Absolute Maximum Rating (Reference to GND) (Note1)

VIN, EN, SW Voltage ..... +20V
Dynamic V_{SW} in 50ns Duration ..... –3V to VIN+3V
BS-SW Voltage ..... +6V
FB Voltage ..... +6V
Junction Temperature Range ..... -40ºC to +150°C
Storage Temperature Range ..... -65ºC to +150°C
Lead Temperature (Soldering 10s) .....+260°C
ESD Classification ..... Class 2

Recommend Operating Conditions (Note2)

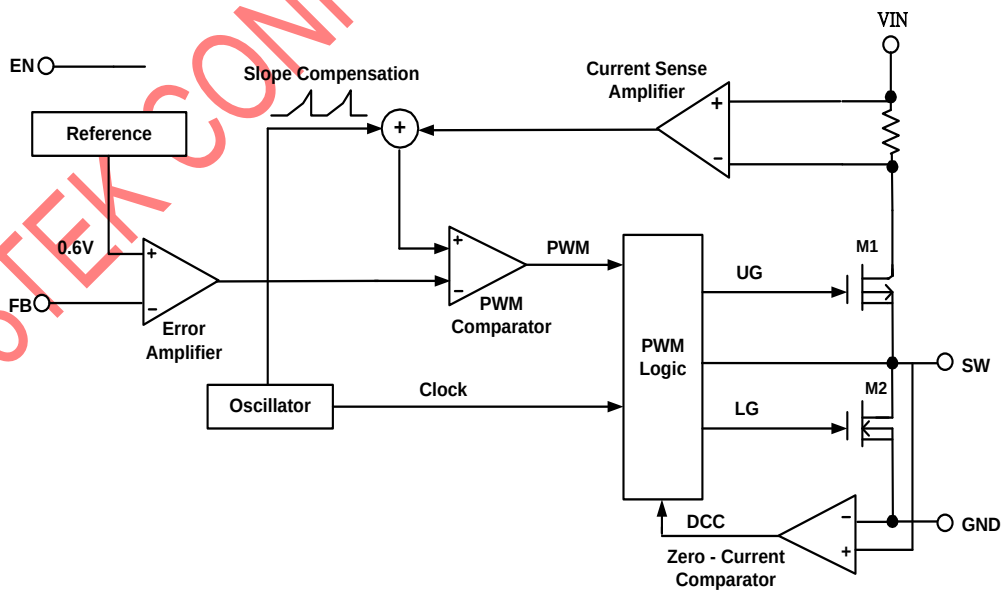
Input Voltage (V_{IN}) ..... +4.5V to +18V
Operating Temperature Range ..... -40°C to +85°C

Thermal information (Note3)

Maximum Power Dissipation(T_A=+25°C) 1.5W
Thermal Resistance (θ_{JA}) ..... 100°C/W
Thermal Resistance (θ_{JC}) .....15°C/W

Note (1): Stress exceeding those listed “Absolute Maximum Ratings” may damage the device.
Note (2): The device is not guaranteed to function outside of the recommended operating conditions.
Note (3): Measured on JESD51-7, 4-Layer PCB.
Note (4): The maximum allowable power dissipation is a function of the maximum junction temperature T_{J_MAX}, the junction to ambient thermal resistance θ_{JA}, and the ambient temperature T_A. The maximum allowable continuous power dissipation at any ambient temperature is calculated by $P_{D_MAX}= (T_{J_MAX}-T_A)/\theta_{JA}$. Exceeding the maximum allowable power dissipation will cause excessive die temperature, and the regulator will go into thermal shutdown. Internal thermal shutdown circuitry protects the device from permanent damage.

Functional Block Diagram



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Electrical Characteristics

Recommended Operating Conditions, T_A = 25°C, V_{IN}=12V, unless otherwise noted.

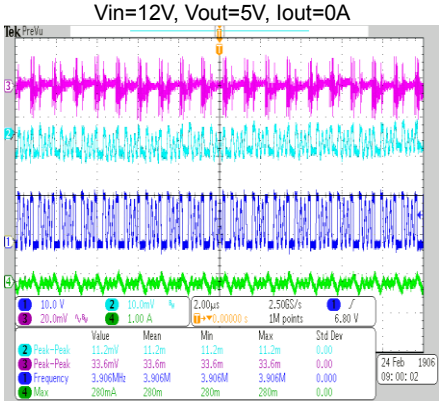
PARAMETER	TEST CONTITIONS	MIN	TYP	MAX	UNIT
Input Supply Voltage		4.5		18	V
Shutdown Current	V_{EN} = 0V		10		μA
Regulated Feedback Voltage V_{REF}	4.5V≦V_{IN}≦18V	0.588	0.6	0.612	V
VIN Under voltage Lockout Threshold			3.75	4.3	V
High Side MOSFET On Resistance			70		mΩ
Low Side MOSFET On Resistance			30		mΩ
Switch Current Limit			5		A
SW Leakage Current	V_{EN}=0V, V_{SW}=0V		1	10	μA
Oscillator Frequency	I_{OUT}=1A		1		MHz
Short Circuit Oscillator Frequency	V_{FB}=0V		280		kHz
Min. On-Time for HS Switch			80		ns
Maximum Duty	V_{FB}≦0.5V		85		%
EN On Threshold		1.3			V
EN Off Threshold				0.5	V
EN Input Current	V_{EN}=1V		1	5	μA
Soft Start Time			0.4		ms
Thermal Shutdown Threshold			160		℃

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TYPICAL PERFORMANCE CHARACTERISTICS

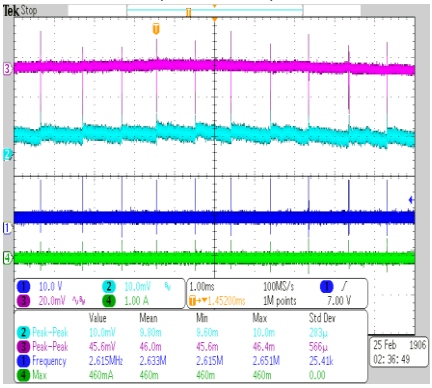
MT3503, C_{IN}=10uF, C_{OUT}=22uF, L=2.2uH(for Vout 5V and Vout 3.3V), L=1uH(for Vout 1V), T_A=25℃

Steady State Test
Vin=12V, Vout=5V, Iout=0A



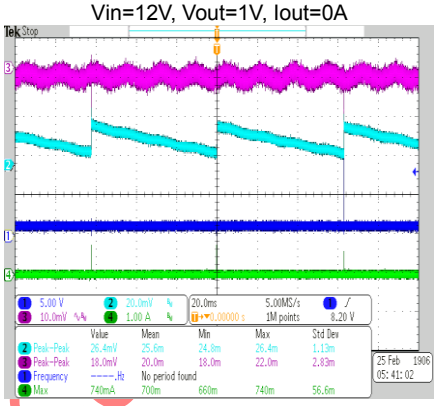
CH1:SW CH2: VOUT CH3: VIN CH4:IL

Steady State Test
Vin=12V, Vout=3.3V, Iout=0A



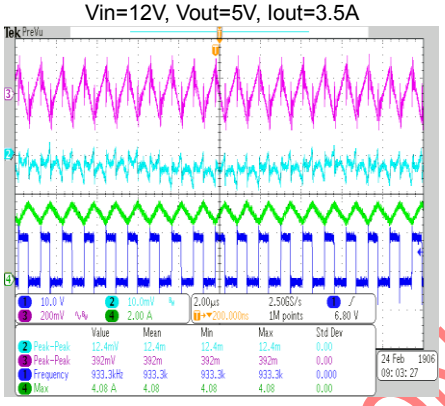
CH1:SW CH2: VOUT CH3: VIN CH4:IL

Steady State Test
Vin=12V, Vout=1V, Iout=0A



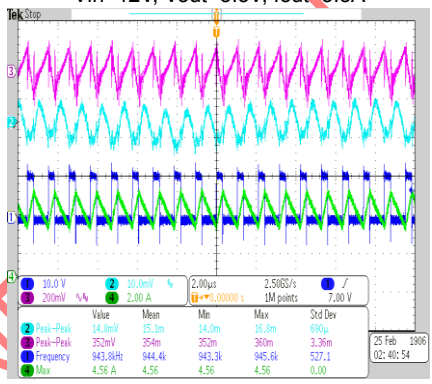
CH1:SW CH2: VOUT CH3: VIN CH4:IL

Steady State Test
Vin=12V, Vout=5V, Iout=3.5A



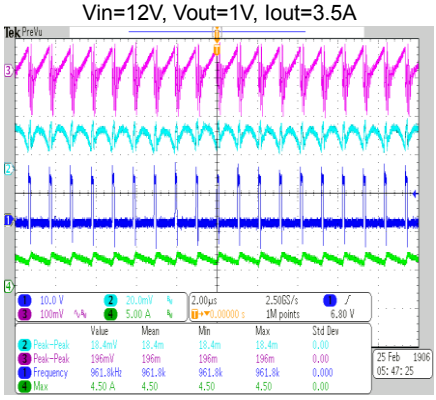
CH1:SW CH2: VOUT CH3: VIN CH4:IL

Steady State Test
Vin=12V, Vout=3.3V, Iout=3.5A



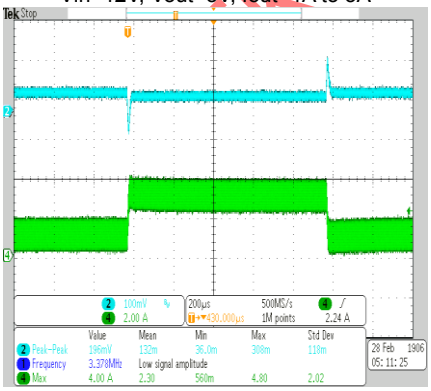
CH1:SW CH2: VOUT CH3: VIN CH4:IL

Steady State Test
Vin=12V, Vout=1V, Iout=3.5A



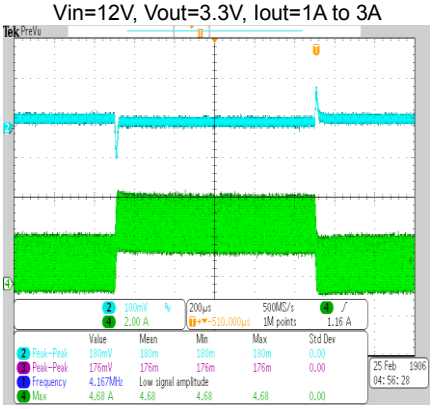
CH1:SW CH2: VOUT CH3: VIN CH4:IL

Load Transient Response
Vin=12V, Vout=5V, Iout=1A to 3A



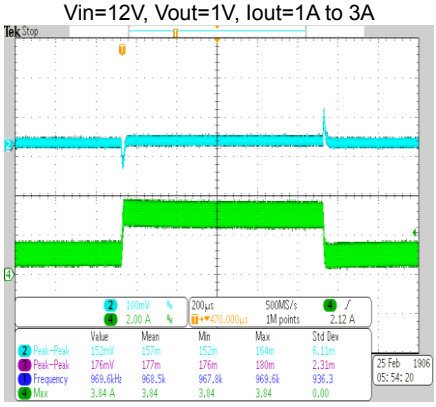
CH2: VOUT CH4:IL

Load Transient Response
Vin=12V, Vout=3.3V, Iout=1A to 3A



CH2: VOUT CH4:IL

Load Transient Response
Vin=12V, Vout=1V, Iout=1A to 3A



CH2: VOUT CH4:IL

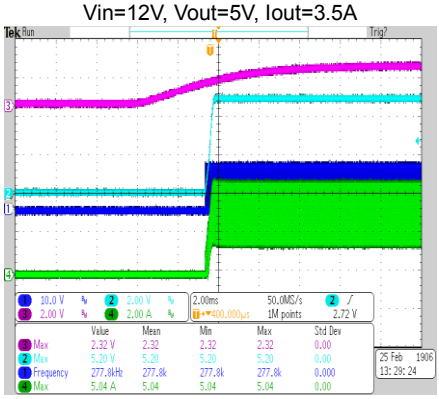
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TYPICAL PERFORMANCE CHARACTERISTICS

MT3503, C_{IN}=10uF, C_{OUT}=22uF, L=2.2uH(for Vout 5V and Vout 3.3V), L=1uH(for Vout 1V), T_A=25℃

EN ON

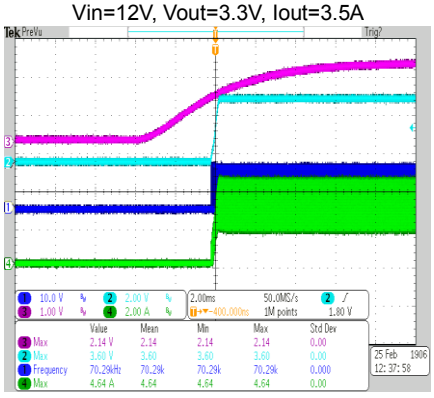
Vin=12V, Vout=5V, Iout=3.5A



CH1:SW CH2: VOUT CH3: EN CH4:IL

EN ON

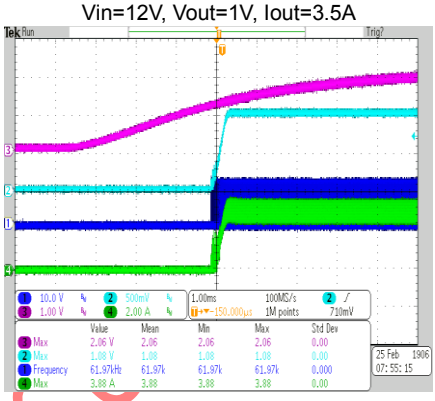
Vin=12V, Vout=3.3V, Iout=3.5A



CH1:SW CH2: VOUT CH3: EN CH4:IL

EN ON

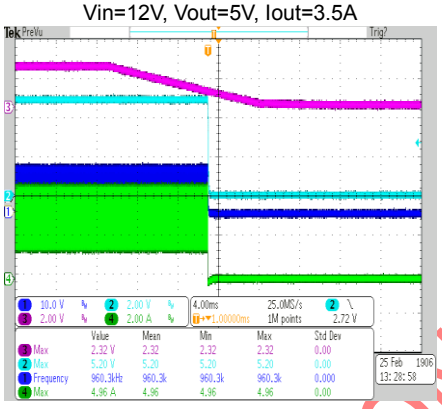
Vin=12V, Vout=1V, Iout=3.5A



CH1:SW CH2: VOUT CH3: EN CH4:IL

EN OFF

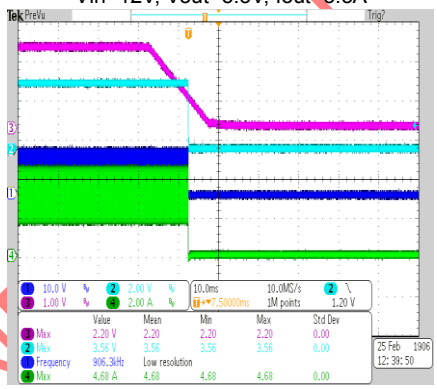
Vin=12V, Vout=5V, Iout=3.5A



CH1:SW CH2: VOUT CH3: EN CH4:IL

EN OFF

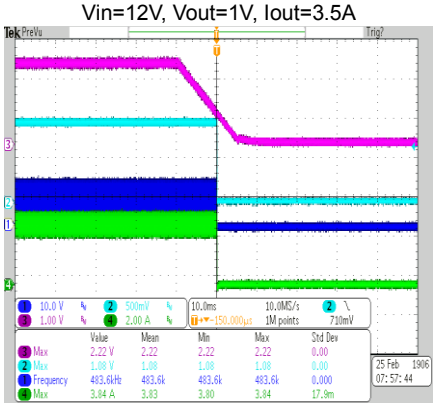
Vin=12V, Vout=3.3V, Iout=3.5A



CH1:SW CH2: VOUT CH3: EN CH4:IL

EN OFF

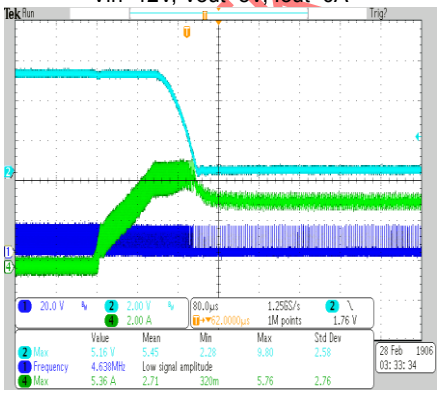
Vin=12V, Vout=1V, Iout=3.5A



CH1:SW CH2: VOUT CH3: EN CH4:IL

Short Circuit Protection

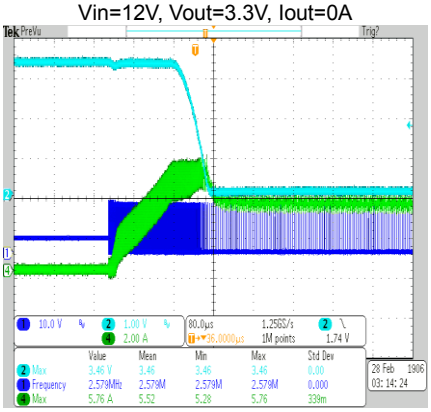
Vin=12V, Vout=5V, Iout=0A



CH1:SW CH2:VOUT CH4:IL

Short Circuit Protection

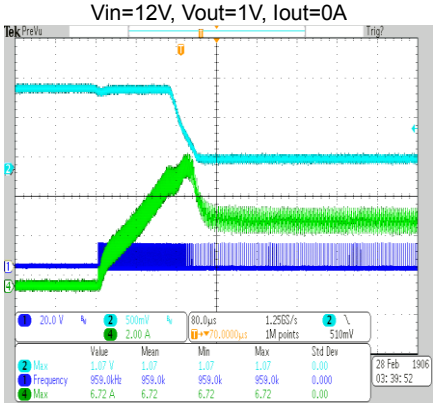
Vin=12V, Vout=3.3V, Iout=0A



CH1:SW CH2:VOUT CH4:IL

Short Circuit Protection

Vin=12V, Vout=1V, Iout=0A



CH1:SW CH2:VOUT CH4:IL

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Theory of Operation

The MT3503/MT3503A is a current mode PWM step-down converter with a constant frequency, and integrates both a main switch and a synchronous rectifier, which provides the high efficiency and eliminates an external Schottky diode.

Current Mode PWM Control

Current mode PWM control provides stable switching and cycle-by-cycle current limit for superior load and line response and protection of the internal main switch and synchronous rectifier. The MT3503/MT3503A switches at a constant frequency and regulates the output voltage. During normal operation, the hidden side is turned on at the beginning of a clock cycle when the VFB voltage is below the Vref (0.6V). The current flows into the inductor and the load increases until the current limit is reached. The switch is turned off and energy stored in the inductor flows through the low side switch into the load until the next clock cycle.

Power Saving Mode

MT3503 power saving mode (PSM) minimizes the switching loss by reducing the switching frequency at light load.

Enable

The EN pin provides ON/OFF control of the regulator. Once the voltage of the EN pin exceeds the threshold voltage, the regulator starts operation and the internal slow start begins to ramp. If the voltage of the EN pin is pulled below the threshold, the regulator will stop switching and reset the internal soft-start. Don't float this pin.

Boost Capacitor

The BST pin and SW pin can be connected by a 100nF low ESR ceramic capacitor, providing the gate drive voltage for the high side MOSFET.

Input Under Voltage Lockout

The MT3503/MT3503A features an Input Under-Voltage Lockout circuit, which shuts down the part when the input voltage drops below failing threshold to prevent unstable operation.

Internal Soft-start

The MT3503/MT3503A features an internal soft−start function, which reduces inrush current and overshoot of the output voltage. Soft−start is achieved by ramping up the reference voltage (V_{ref}) which is applied to the input of the error amplifier. The typical soft−start time, depending on the component's values on AP circuit.

Short Circuit Protection

The MT3503/MT3503A has short circuit protection. When the output is shorted to ground, the oscillator's frequency is reduced to prevent the inductor's current from increasing beyond the high side MOSFET current limit. The frequency will return to the normal values once the short circuit condition is removed and the feedback voltage > 0.25V.

Maximum Load Current

The MT3503/MT3503A can operate down to 4.5V input voltage; however the maximum load current decreases at lower input due to large IR drop on the main switch and synchronous rectifier. The slope compensation signal also

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reduces the inductor's peak current as a function of the duty cycle to prevent sub-harmonic oscillations at duty cycles greater than 50%.

Application Information

Input Capacitor Selection

The input capacitor must sustain the ripple current produced during the period of "ON" state of the high side MOSFET, so a low ESR ceramic capacitor is required to minimize the loss. The input ripple current RMS value can be calculated by the following equation:

$$I_{INRMS} = I_{OUT}\sqrt{D\times(1-D)}$$

Where D is the duty cycle, I_{INRMS} is the input RMS current, and I_{OUT} is the load current. The equation reaches its maximum value with D = 0.5. The loss of the input capacitor can be calculated by the following equation:

$$P_{CIN} = ESR_{CIN}\times I_{INRMS}{}^2$$

Where P_{CIN} is the power loss of the input capacitor and ESR_{CIN} is the effective series resistance of the input capacitance. Due to large di/dt through the input capacitor, electrolytic or ceramics should be used.

Inductor Selection

The inductor selection is to meet the requirements of the output voltage ripple and affects the load transient response. The higher inductance can reduce the inductor's ripple current and induce the lower output ripple voltage. The ripple voltage and current are approximated by the following equations:

$$\Delta I = \frac{V_{IN}-V_{OUT}}{F_S\times L}\bullet\frac{V_{OUT}}{V_{IN}}$$

$$\Delta V_{OUT} = \Delta I\times ESR$$

Although the increase of the inductance reduces the ripple current and voltage, it contributes to the decrease of the response time for the regulator to load transient. The way to set a proper inductor value is to have the ripple current (△I) be approximately 20%~50% of the maximum output current. Once the value has been determined, select an inductor capable of carrying the required peak current without going into saturation. It is also important to have the inductance tolerance specified to keep the accuracy of the system. 20% tolerance (at room temperature) is reasonable for the most inductor manufacturers. For some types of inductors, especially those with ferrite core, the ripple current will increase abruptly when it saturates, which will result in larger output ripple voltage.

Output Capacitor Selection

An output capacitor is required to filter the output and supply the load transient current. The high capacitor value and low ESR will reduce the output ripple and the load transient drop. In typical switching regulator design, the ESR of the output capacitor bank dominates the transient response. The number of output capacitors can be determined by the following equations:

$$ESR_{MAX} = \frac{\Delta V_{ESR}}{\Delta I_{OUT}}$$

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$$\text{Number of capacitors} = \frac{ESR_{CAP}}{ESR_{MAX}}$$

$\triangle V_{ESR}$ = change in output voltage due to ESR

$\triangle I_{OUT}$ = load transient

ESR_{CAP} = maximum ESR per capacitor (specified in manufacturer's data sheet)

ESR_{MAX} = maximum allowable ESR

High frequency decoupling capacitors should be placed as closely to the power pins of the load as physically possible. For the decoupling requirements, please consult the capacitor manufacturers for confirmation.

Output Voltage

The output voltage is set using the FB pin and a resistor divider connected to the output as shown in AP Circuit below. The output voltage (V_{OUT}) can be calculated according to the voltage of the FB pin (V_{FB}) and ratio of the feedback resistors by the following equation, where (V_{FB}) is 0.6V:

$$V_{FB} = V_{OUT} \times \frac{R_2}{(R_1 + R_2)}$$

Thus the output voltage is:

$$V_{OUT} = 0.6 \times \frac{(R_1 + R_2)}{R_2}$$

Choose R1=100kΩ~200kΩ to ensure feedback loop noise immunity. It is optional to add a feed-forward capacitor C_{FF}=22~33pF in parallel with R1 to achieve better transient response performance.

Layout Consideration

The physical design of the PCB is the final stage in the design of power converter. The improperly designed PCB could radiate excessive EMI and contribute instability to the power converter. Follow the PCB layout guidelines below to ensure better performance of MT3503/MT3503A.

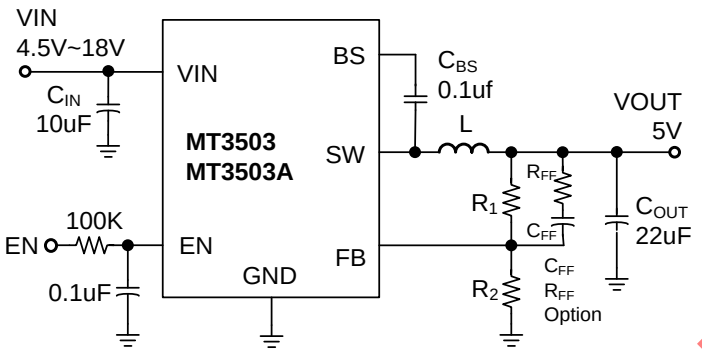
(1). The loop (Vin-SW-L-Cout-GND) indicates a high current path. The traces within the loop should be kept as wide and short as possible to reduce parasitic inductance and high-frequency loop area. It is also good for efficiency improvement.

(2). Place Input capacitor as close as possible to the IC Pins (Vin and GND) and the input loop area should be as small as possible to reduce parasitic inductance, input voltage spike and noise emission.

(3). Feedback components (R_1, R_2 and C_{FF}) should be routed as far away from the inductor and the BS and SW pins to minimize noise and EMI issue.

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MT3503/MT3503A Application Schematic



EVB BOM List

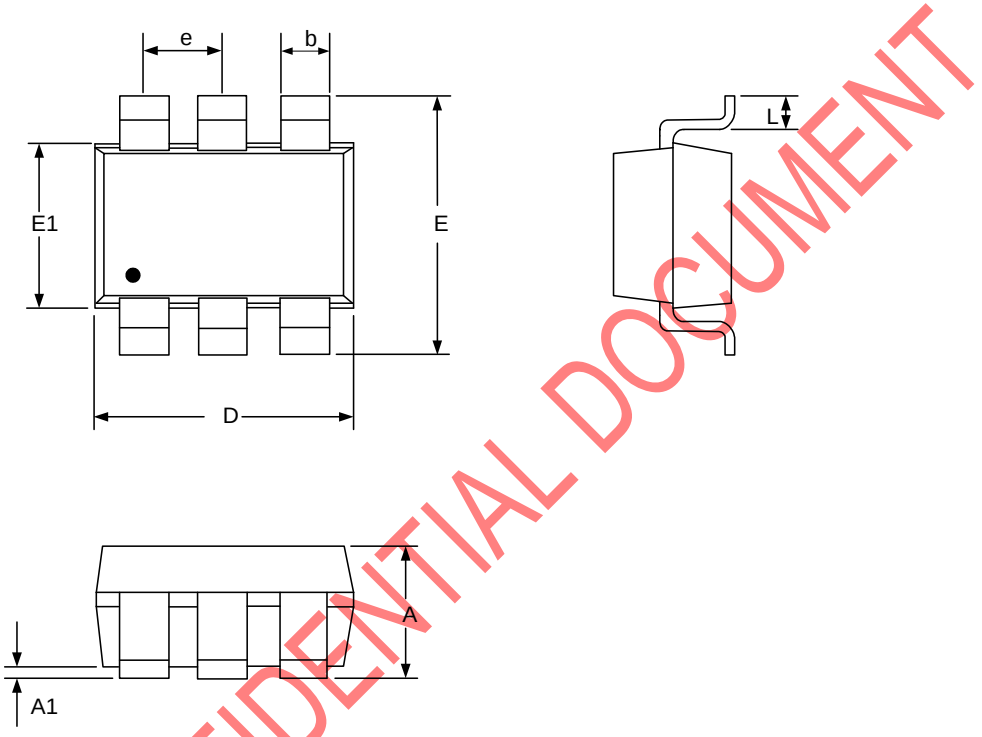
Qty	Ref	Value		Description	Package
1	C_{IN}	10μF		Ceramic Capacitor, 25V, X5R	0805
1	C_{OUT}	22μF		Ceramic Capacitor, 25V, X5R	0805
2	C_{EN} , C_{BS}	0.1μF		Ceramic Capacitor, 16V, X5R	0603
NA	C_{FF} R_{FF}	Option		Ceramic Capacitor, 16V, X5R	0603
1	L	Vout=5V	2.2μH	Inductor,GSTM4020P 18mΩ, 6A	SMD
		Vout=3.3V	2.2μH		
		Vout=2.5V	2.2μH		
		Vout=1.8V	1.5μH		
		Vout=1.5V	1.5μH		
		Vout=1.2V	1μH		
		Vout=1V	1μH		
1	R1	Vout=5V	100KΩ	Resistor, ±1%	0603
		Vout=3.3V	100KΩ		
		Vout=2.5V	100KΩ		
		Vout=1.8V	100KΩ		
		Vout=1.5V	100KΩ		
		Vout=1.2V	100KΩ		
		Vout=1V	100KΩ		
1	R2	Vout=5V	13.7KΩ	Resistor, ±1%	0603
		Vout=3.3V	22.1KΩ		
		Vout=2.5V	31.6KΩ		
		Vout=1.8V	49.9KΩ		
		Vout=1.5V	66.5KΩ		
		Vout=1.2V	100KΩ		
		Vout=1V	150KΩ		
1	R_{EN}	10K~100KΩ		Resistor, ±1%	0603
1	Power IC	MT3503/MT3503A		Step-Down DC/DC Converter	TSOT23_6L

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Package Information

TSOT23_6L Outline Dimensions

Unit: inches/mm

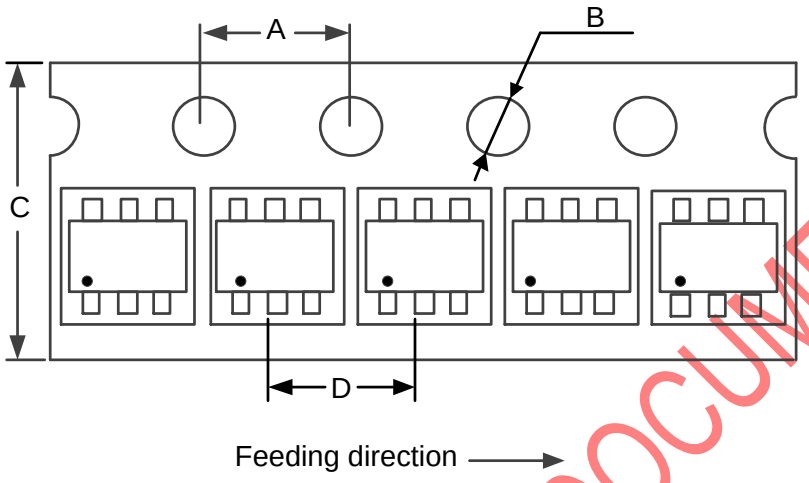


SYMBOLS	MILLIMETERS		INCHES	
	MIN.	MAX.	MIN.	MAX.
A	0.70	1.1	0.027	0.043
A1	0.00	0.1	0.000	0.004
b	0.30	0.50	0.012	0.020
D	2.70	3.10	0.106	0.122
E1	1.40	1.80	0.055	0.071
e	0.84	1.04	0.033	0.041
E	2.60	3.00	0.102	0.118
L	0.30	0.60	0.012	0.024

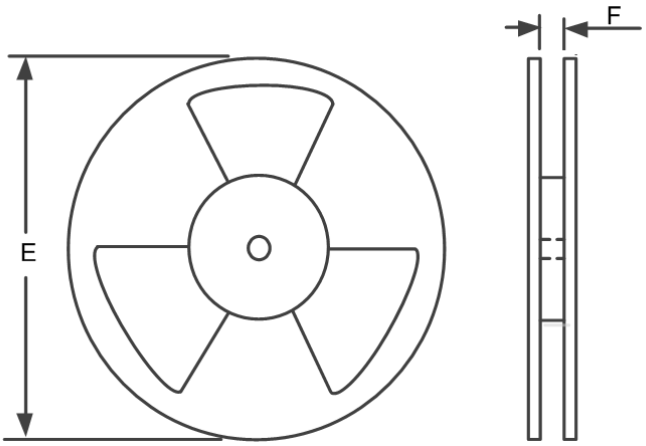
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Tape & Reel Carrier Dimensions

1. Orientation / Carrier Tape Information :



2. Rokreel Information :



3. Dimension Details :

PKG Type	A	B	C	D	E	F	Q'ty/Reel
TSOT23_6L	4.0 mm	1.5 mm	8.0 mm	4.0 mm	7 inches	9.0 mm	3,000

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Reflow Profile

Classification Of IR Reflow Profile

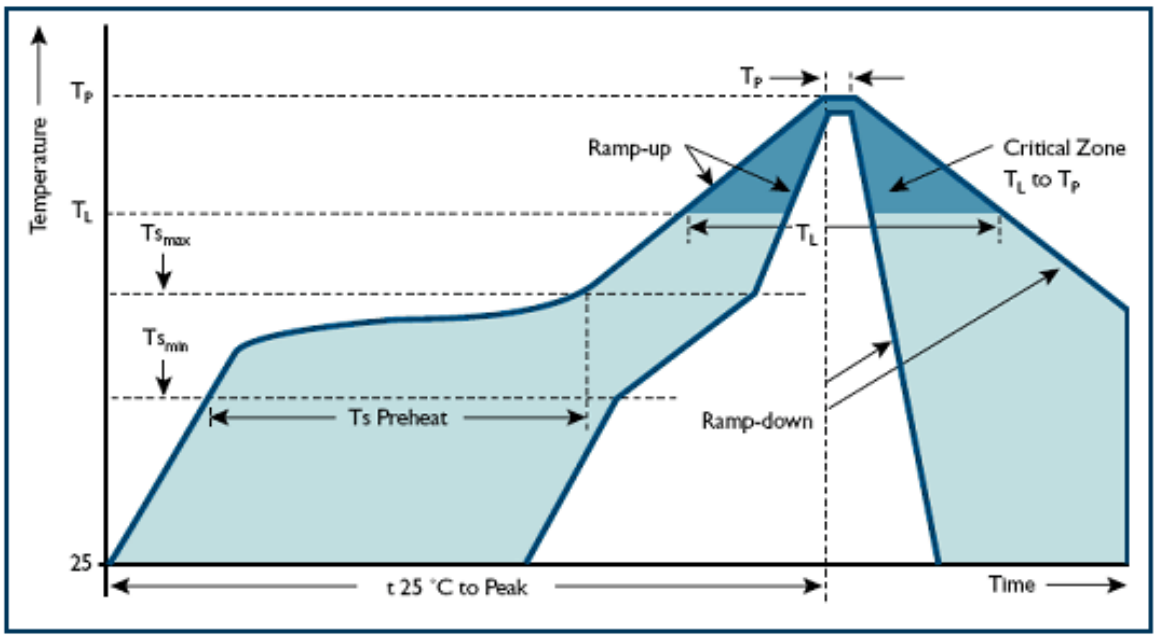
Reflow Profile	Green Assembly
Average Ramp-Up Rate (Ts_{min} to Tp)	1~2℃/second, 3℃/second max.
Preheat & Soak -Temperature Min(Ts_{min}) -Temperature Max(Ts_{max}) -Time(ts_{min} to ts ts_{max})	 150℃ 200℃ 60~120 seconds
Time maintained above: -Temperature(T_L) -Time(t_L)	 217℃ 60~150 seconds
Peak Temperature(Tp)	See Classification Temp in table 1
Time within 5℃ of actual Peak Temperature(tp)	30 seconds max.
Ramp-Down Rate	6℃/second max.
Time 25℃ to Peak Temperature	8 minutes max.

* Tolerance for peak profile Temperature (T_p) is defined as a supplier minimum and a user maximum.
** Tolerance for time at peak profile temperature (t_p) is defined as a supplier minimum and a user maximum.

Table 1. Pb-free Process – Classification Temperatures (Tc)

Package Thickness	Volume mm^3 <350	Volume mm^3 350-2000	Volume mm^3 >2000
<1.6 mm	260 ℃	260 ℃	260 ℃
1.6 mm – 2.5 mm	260 ℃	250 ℃	245 ℃
≥2.5 mm	250 ℃	245 ℃	245 ℃

Note: For all temperature information, please refer to topside of the package, measured on the package body surface.



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